

CONSTANTS	$F_s = 32,000$ psi	allowable steel stress, Grade 60 (§8.3.3.1)
	$F_b = 0.45f'_m$	allowable flexural compression, reinforced (§8.3.4.2.2)
	$E_m = 900f'_m$	concrete masonry modulus; clay: $700f'_m$ (§4.2.2)
	$E_s = 29,000,000$ psi	steel modulus (§4.2.2)

1 SHEAR (ASD)

WHERE: TMS 402-16 §8.3.5 · ANSWERS: STEEL-ONLY / MASONRY-ONLY / COMBINED SHEAR CAPACITY · APPLIES: BEAMS, WALLS IN-PLANE, PIERS

$$f_v = \frac{V}{A_{nv}} \leq F_v = (F_{vm} + F_{vs}) \gamma_g \quad (1)$$

f_v	unit shear stress
A_{nv}	net shear area
F_v	total allowable unit shear stress
F_{vm}	masonry share
F_{vs}	shear-reinforcement share
γ_g	0.75 partially grouted shear wall, else 1.0

$$F_{vm} = \frac{1}{2} \left[4.0 - 1.75 \frac{M}{Vd_v} \right] \sqrt{f'_m} + 0.25 \frac{P}{A_{nv}} \quad (2)$$

d_v	actual member depth in direction of shear
$M/(Vd_v)$	taken ≤ 1.0 ; axial term P zero for beams

$$F_{vs} = 0.5 \frac{A_v F_s d_v}{A_{nv} s} \quad (3)$$

A_v shear reinforcement area per spacing s

LIMITS

$M/(Vd_v) \leq 0.25$	$F_v \leq 3\sqrt{f'_m} \gamma_g$
$M/(Vd_v) \geq 1.0$	$F_v \leq 2\sqrt{f'_m} \gamma_g$
between	interpolate
special reinforced shear wall	F_{vm} coefficient $\frac{1}{2} \rightarrow \frac{1}{4}$; design for $1.5\times$ seismic V

2 FLEXURE (ASD)

WHERE: TMS 402-16 §8.3.4 · ANSWERS: BEAM / WALL MOMENT CAPACITY · APPLIES: BEAMS, LINTELS, WALLS OUT-OF-PLANE

$$n = \frac{E_s}{E_m} \quad \rho = \frac{A_s}{bd} \quad k = \sqrt{(n\rho)^2 + 2n\rho} - n\rho \quad j = 1 - \frac{k}{3} \quad (4)$$

n	modular ratio
k	neutral-axis depth ratio
j	moment-arm ratio

$$M \leq \min \left(A_s F_s j d, \frac{1}{2} F_b k j b d^2 \right) \quad (5)$$

$A_s F_s j d$	steel governs
$\frac{1}{2} F_b k j b d^2$	masonry governs

3 AXIAL + INTERACTION (ASD)

WHERE: TMS 402-16 §8.2.4 (UNREINFORCED) · ANSWERS: UNREINFORCED WALL AXIAL CAPACITY, P+M UNITY CHECK

$$F_a = \frac{1}{4} f'_m \left[1 - \left(\frac{h}{140r} \right)^2 \right] \quad \left(\frac{h}{r} \leq 99 \right) \quad F_a = \frac{1}{4} f'_m \left(\frac{70r}{h} \right)^2 \quad \left(\frac{h}{r} > 99 \right) \quad (6)$$

F_a	allowable axial compressive stress
h	effective height
r	radius of gyration

$$\frac{f_a}{F_a} + \frac{f_b}{F_b} \leq 1 \quad P \leq \frac{1}{4}P_e = \frac{1}{4} \frac{\pi^2 E_m I_n}{h^2} \left(1 - 0.577 \frac{e}{r}\right)^3 \quad (7)$$

f_a, f_b computed axial, flexural stresses
 F_b here $\frac{1}{3}f'_m$ unreinforced (§8.2.4.2), not $0.45f'_m$
 P_e Euler buckling load; unreinforced check only
 I_n net moment of inertia
 e eccentricity

4 SHEAR + FLEXURE (STRENGTH DESIGN)

WHERE: TMS 402-16 §9.3 · ANSWERS: ϕV_n SHEAR / ϕM_n MOMENT (SD) · ϕ : SHEAR 0.80, FLEXURE 0.90 (§9.1.4)

$$V_u \leq \phi V_n = \phi (V_{nm} + V_{ns}) \gamma_g \quad (8)$$

V_{nm}, V_{ns} nominal masonry, reinforcement shares

$$V_{nm} = \left[4.0 - 1.75 \frac{M_u}{V_u d_v}\right] A_{nv} \sqrt{f'_m} + 0.25 P_u \quad V_{ns} = 0.5 \frac{A_v}{s} f_y d_v \quad (9)$$

$M_u/(V_u d_v)$ taken ≤ 1.0

$$a = \frac{A_s f_y + P_u}{0.80 f'_m b} \quad M_n = (A_s f_y + P_u) \left(d - \frac{a}{2}\right) \quad (10)$$

a equivalent stress-block depth ($0.80f'_m$, §9.3.2)
 ϵ_{mu} 0.0025 CMU, 0.0035 clay (§9.3.2)

LIMITS

$M_u/(V_u d_v) \leq 0.25$ $V_n \leq 6A_{nv} \sqrt{f'_m} \gamma_g$
 $M_u/(V_u d_v) \geq 1.0$ $V_n \leq 4A_{nv} \sqrt{f'_m} \gamma_g$
 between interpolate
 special wall capacity design shear $\geq$ shear at $1.25M_n$, need not exceed $2.5V_u$ (§7.3.2.6.1)

5 ANCHOR BOLTS (ASD)

WHERE: TMS 402-16 §8.1.3 · ANSWERS: HEADED-BOLT TENSION / SHEAR / COMBINED CAPACITY · MINIMUM MODE GOVERNS · BENT-BAR: ADD PULLOUT (EQ 8-3)

$$B_a = \min \left(B_{ab} = 1.25 A_{pt} \sqrt{f'_m}, \quad B_{as} = 0.6 A_b f_y \right) \quad (11)$$

B_a allowable tension
 B_{ab} masonry breakout
 B_{as} steel yield
 A_{pt} projected tension breakout area
 A_b bolt cross-sectional area

$$B_v = \min \left(B_{vb} = 1.25 A_{pv} \sqrt{f'_m}, \quad B_{vc} = 350 \sqrt[4]{f'_m A_b}, \quad B_{vpry} = 2.0 B_{ab}, \quad B_{vs} = 0.36 A_b f_y \right) \quad (12)$$

B_v allowable shear
 B_{vb} masonry breakout
 B_{vc} masonry crushing
 B_{vpry} pryout
 B_{vs} steel yield
 A_{pv} projected shear breakout area

$$\frac{b_a}{B_a} + \frac{b_v}{B_v} \leq 1 \quad (13)$$

b_a, b_v applied tension, shear